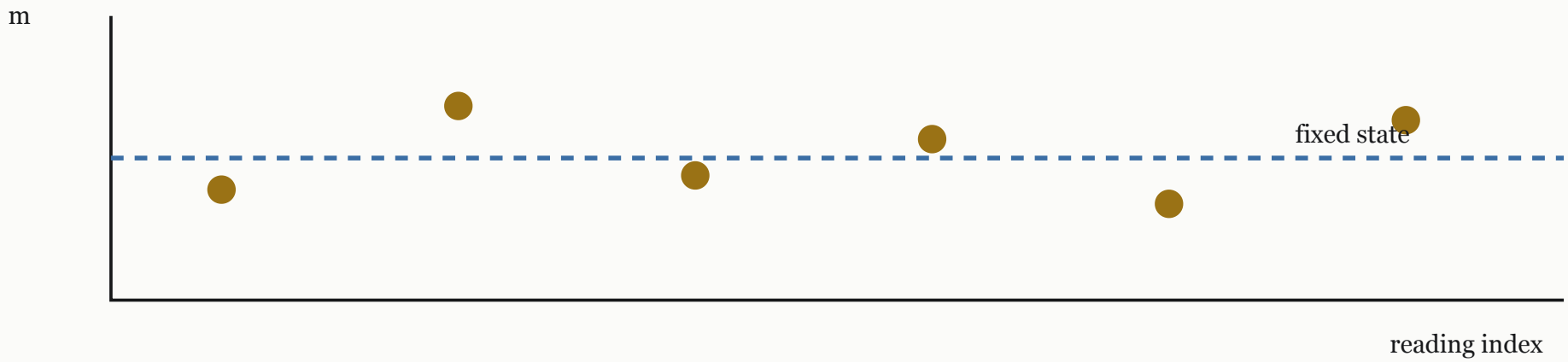
WHY

Uncertainty

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

Report an estimate and the uncertainty around it.

WHY

Uncertainty

What does a distribution add to a model?

$$z = x + \varepsilon$$

- x unknown state (m)
- z observed reading (m)
- ε random error (m)

Deterministic prediction

One output for specified inputs.

Probabilistic description

Possible outcomes and their weights.

> Uncertainty describes the model, not a moving wall.

Random variables

How does an outcome become a number?

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

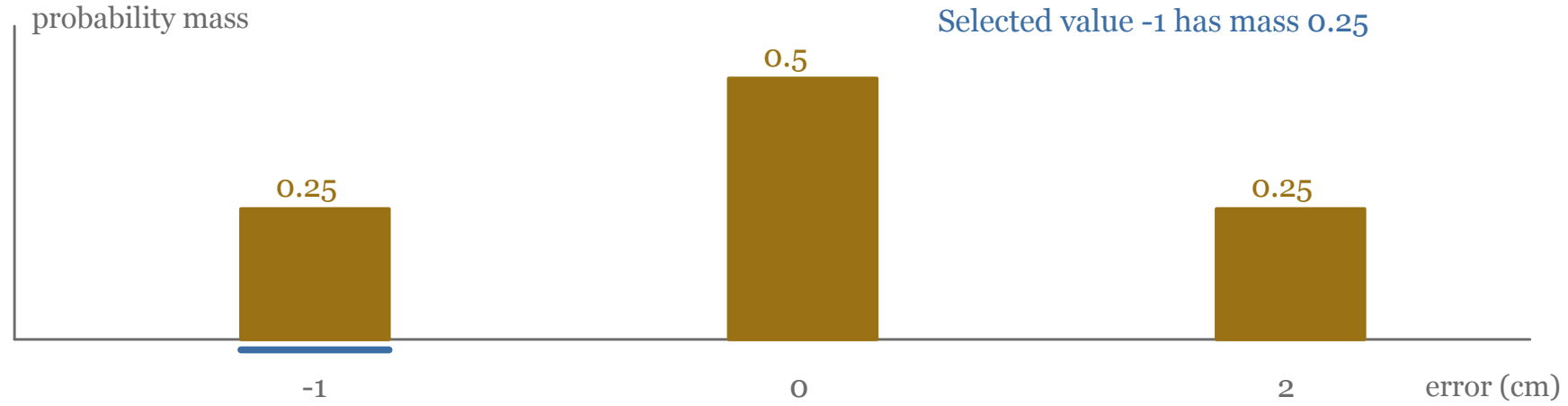
- Ω possible outcomes
- ω one outcome
- x random variable: numerical error (cm)

Outcome A → −1 cm	Outcome B → 0 cm	Outcome C → 2 cm

> The rule is fixed; the outcome is uncertain.

PMF and PDF

How do probabilities attach to discrete values?



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

- x discrete random error
- a possible value (cm)
- $p(a)$ probability mass
- P event probability

DEFINITION AND REASONING

PMF and PDF

What does the three-point error model predict?

WORKED EXAMPLE E1

Error values −1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- P event probability
- x error (cm)
- μ mean error (cm)

CHOOSE, EXPLAIN, VOTE

PMF and PDF

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

PMF and PDF

Why does a continuous event need an area?

$$P(a \le x \le b) = \int_a^b p(t)\,dt$$

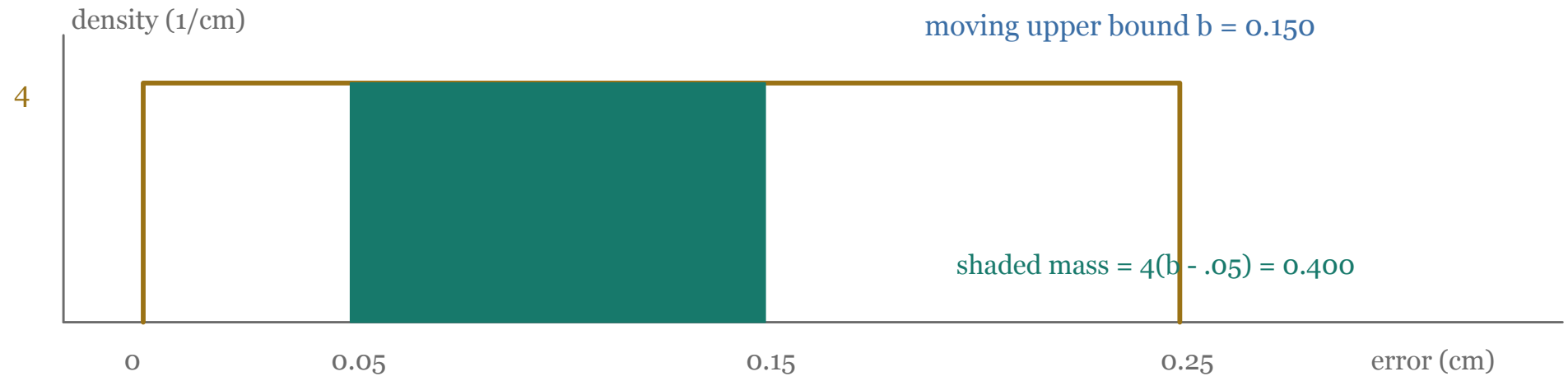
- x continuous random quantity
- a, b interval endpoints
- t integration variable
- $p(t)$ density, inverse units of x

$$P(x = c) = \int_c^c p(t)\,dt = 0$$

- c one fixed value
- P probability for a variable with a density

PMF and PDF

Can a valid density be greater than one?



WORKED EXAMPLE E2 · UNIFORM ERROR

$$p(t) = 4 \;\; (0 \le t \le .25), \qquad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

- t, x error (cm)
- p density (cm^{-1})
- P interval probability

CHOOSE, EXPLAIN, VOTE

PMF and PDF

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is .2, the interval width alone.

Moments

How do we average a function of a random value?

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete random variable
- a support value
- g function being averaged
- p PMF

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous random variable
- t integration variable
- p density
- g assume E[|g(x)|] is finite

CHOOSE, EXPLAIN, VOTE

Moments

What does centering remove?

For an error x, $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2. Which variance is correct?

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean.

DEFINITION AND REASONING

Moments

What is the mean of a random column vector?

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x random column in $\mathbb{R}^2$
- x_i component i
- μ mean column in $\mathbb{R}^2$

> Average corresponding components; preserve the column shape.

Samples use superscripts: $x^{(s)}$. Components use subscripts: x_i.

DEFINITION AND REASONING

Moments

Why does covariance use an outer product?

$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

- x, μ — random column and mean in R^d
- Σ — population covariance, d × d
- i, j — coordinate indices

> Diagonal: component variance. Off-diagonal: paired deviations.

DEFINITION AND REASONING

Moments

Why is covariance positive semidefinite?

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed direction column in $\mathbb{R}^d$
- x, μ random column and population mean
- Σ population covariance

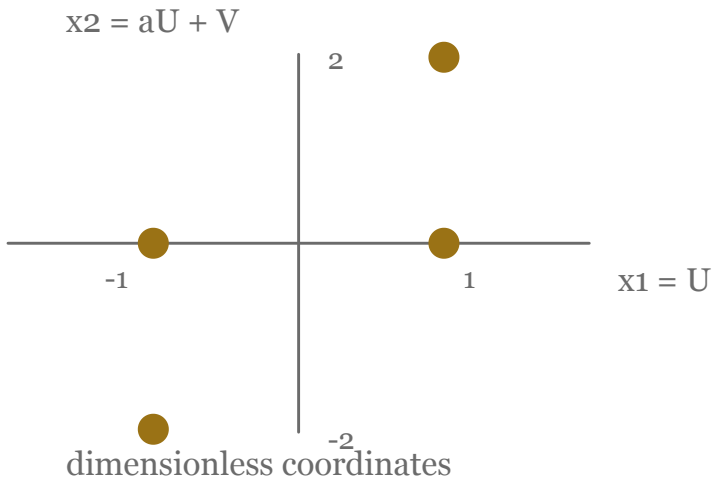
$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C sample covariance estimate
- X_c N centered sample rows
- N number of samples

DEFINITION AND REASONING

Moments

How do independent signs create correlation?



a = 1.00

four equally likely outcomes

a: 0 to 1 adds the shared sign

WORKED EXAMPLE E4

Independent fair signs U,V. Let $x = (U, U+V)^T$.

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

U, V independent, equally likely ± 1

x, μ random column and mean

Σ population covariance

CHOOSE, EXPLAIN, VOTE

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

A 2: both components share U.

B 0: independent sources imply independent outputs.

C 1: doubling a coefficient has no effect.

DEFINITION AND REASONING

Moments

Does zero covariance guarantee independence?

WORKED EXAMPLE E5

x is uniform on {−1,0,1}; y=x². Is y independent of x?

$$\operatorname{Cov}(x, y) = 0, \qquad P(y = 0 \mid x = 0) = 1 \neq \tfrac{1}{3} = P(y = 0)$$

x, y	random scalars
P	event probability
Cov	population covariance

Gaussian geometry

What defines a multivariate Gaussian density?

$$p(x) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp(-d_M^2/2)$$

- x, μ random column and mean in R^{d}
- d dimension
- $\Sigma \succ 0$ positive definite covariance
- $|\Sigma|$ determinant
- p density
- d_M^2 squared Mahalanobis distance

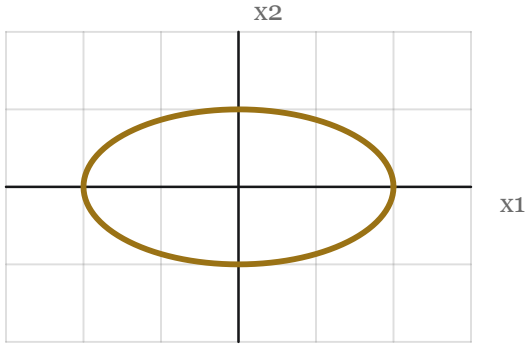
$$d_M^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

- d_M Mahalanobis distance
- Σ^{-1} inverse covariance
- $x - \mu$ centered displacement

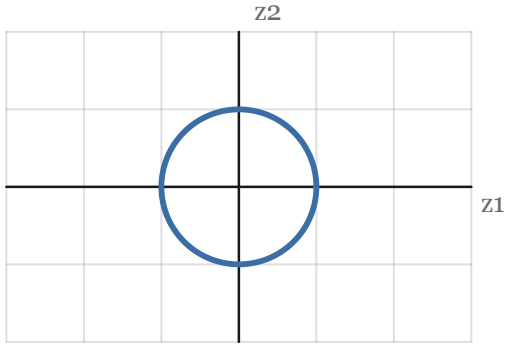
Gaussian geometry

Rotate, then standardize each coordinate

centered original coordinates



divide x1 by its standard deviation

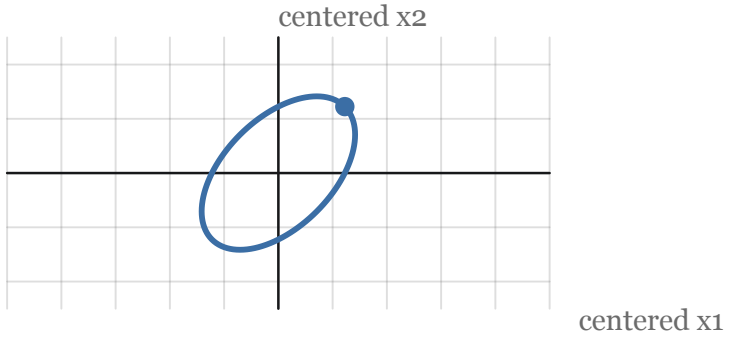


$$\Sigma = Q\Lambda Q^T, \quad z = \Lambda^{-1/2} Q^T (x - \mu)$$

Symbols Q eigenvectors; Λ positive eigenvalues; z whitened displacement; x, μ position and mean.

Gaussian geometry

A contour radius becomes an ellipse axis



unit Mahalanobis contour

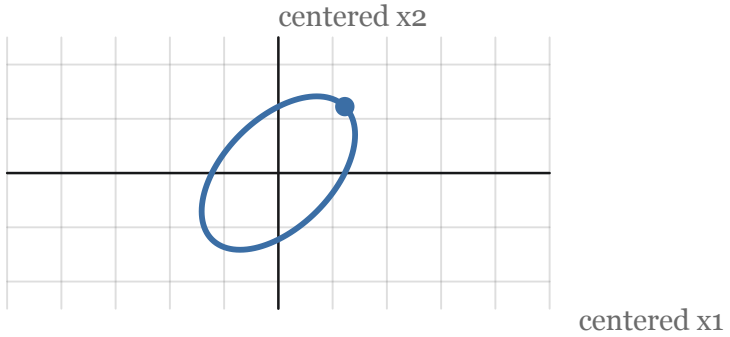
eigenvalues 3 and 1

$$x = \mu + Q\Lambda^{1/2} z, \qquad \|z\| = c$$

Symbols c standardized radius; $\sqrt{\lambda_i}$ standard deviation along eigenvector q_i.

Gaussian geometry

Equal lengths can have unequal surprise



unit Mahalanobis contour

eigenvalues 3 and 1

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Symbols Σ covariance; μ mean; compare $x = (2, 0)^T$ and $y = (2, -2)^T$.

WORKED EXAMPLE

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

Result d_M covariance-scaled distance.

CHOOSE, EXPLAIN, VOTE

Gaussian geometry

Which direction makes a point less surprising?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

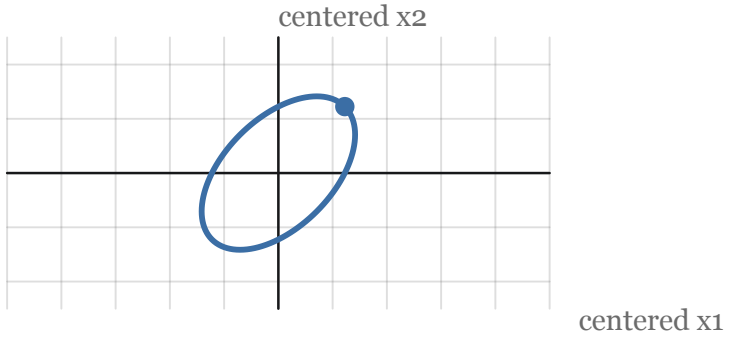
A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

C $p(u) < p(v)$: higher variance always lowers density.

Gaussian geometry

A unit ellipse contains less than half



unit Mahalanobis contour

eigenvalues 3 and 1

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

Symbols c contour radius; $d = 2$ dimension; the formula assumes a nonsingular Gaussian.

WORKED EXAMPLE

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

Result $c_{.95}$ radius enclosing 95%.

Bayes

Which population does a positive select?

all model trials: 10,000



event 100 non-event 9,900

Conditioning selects a new denominator

$$P(H \mid +) \text{ need not equal } P(+ \mid H)$$

Symbols H event of interest; $+$ positive sensor result; the bar restricts the population.

Bayes

Write the joint event in two ways

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

Symbols $P(H)$ prior; $P(+ \mid H)$ likelihood; $P(+) > 0$ evidence.

Bayes

Normalize over both sources of positives

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

Symbols $\neg H$ complement; the two joint events are disjoint and exhaustive.

Bayes

Rare events can lose the positive majority

all model trials: 10,000



event 100 non-event 9,900

Conditioning selects a new denominator

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

Symbols 10,000 model trials: 100 event cases and 9,900 non-event cases.

WORKED EXAMPLE

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

Result H event; $+$ positive reading.

CHOOSE, EXPLAIN, VOTE

Bayes

Does a better base rate change the sensor?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

A .9: sensitivity is the positive predictive value.

B .1: an observation cannot change the prior.

C .5: true and false positives contribute equally.

Break

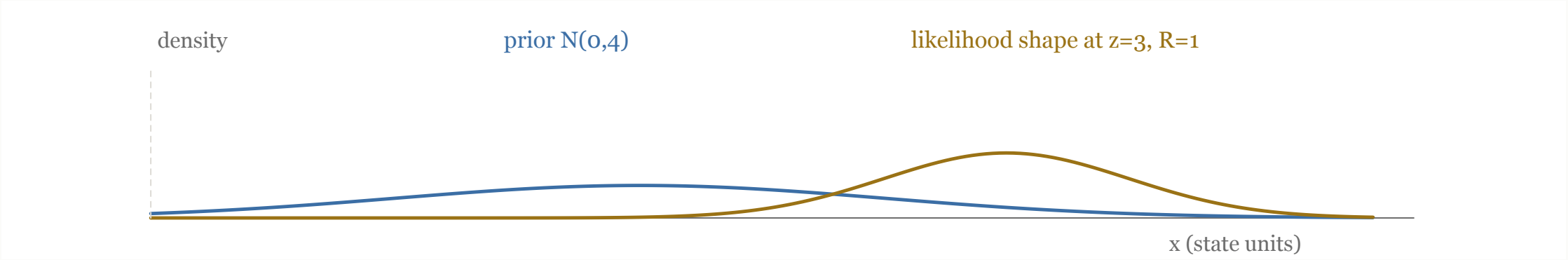
Pause: three different kinds of probability

$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

Symbols A event; p density; conditioning changes the reference population.

Gaussian conditioning

A numeric reading updates a curve



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

Symbols x uncertain state; z observed value; $P_0, R > 0$ variances; independent noise.

Gaussian conditioning

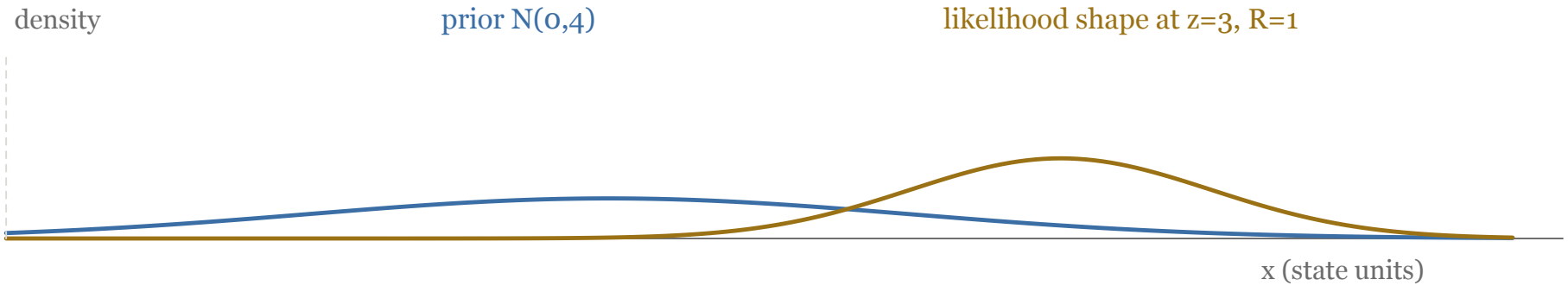
Add precisions, then complete the square

$$P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(P_0^{-1}\mu_0 + R^{-1}z)$$

Symbols P_1, μ_1 posterior variance and mean; inverse variance is precision.

Gaussian conditioning

The gain weights the innovation

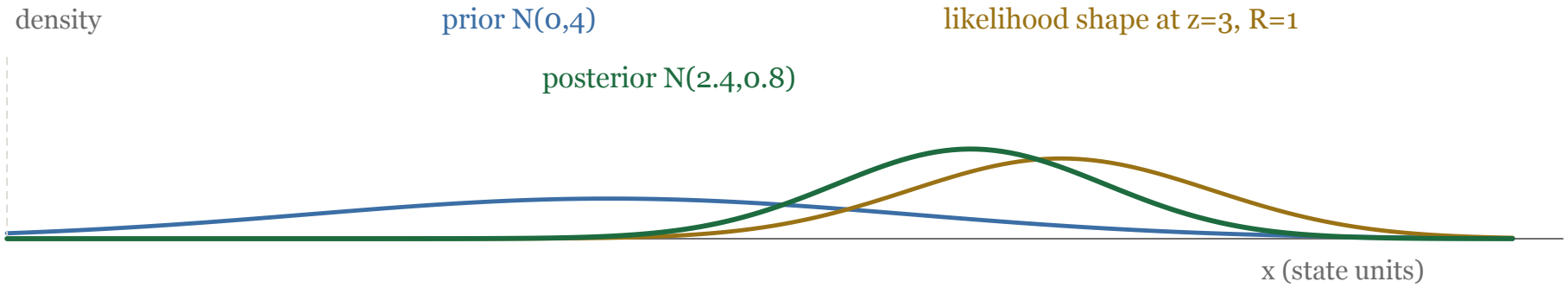


$$K = \frac{P_0}{P_0 + R}, \quad \mu_1 = \mu_0 + K(z - \mu_0), \quad P_1 = (1 - K)P_0$$

Symbols K gain; $z - \mu_0$ innovation; $0 < K < 1$ for positive variances.

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

Symbols P_0, R are variances, not standard deviations.

WORKED EXAMPLE

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

Result K gain; μ_1, P_1 posterior mean, variance.

CHOOSE, EXPLAIN, VOTE

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$, reading $z = 3$. Noise variance rises from $R = 1$ to $R = 4$. What changes?

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning

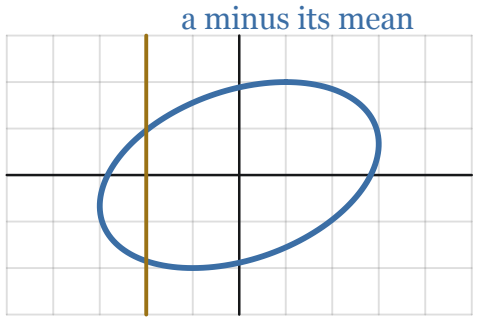
A correlated block predicts another block

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

Symbols Jointly Gaussian blocks; invertible Σ_{bb}. $a \in \mathbb{R}^p$, $b \in \mathbb{R}^q$; $\Sigma_{ab} : p \times q$.

Gaussian conditioning

Conditioning removes predictable variance



slice fixes b; a remains uncertain

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

Symbols Σ_{aa} marginal covariance; $\Sigma_{ba} = \Sigma_{ab}^{T}$; joint Gaussian, invertible Σ_{bb}.

CHOOSE, EXPLAIN, VOTE

Gaussian conditioning

Does the slice width change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Estimation

Choose a parameter to explain fixed data

$$b = A\theta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Symbols $b \in \mathbb{R}^N$ data; $A \in \mathbb{R}^{N \times p}$ known; $\theta \in \mathbb{R}^p$ unknown; σ^2 known.

Estimation

Gaussian products become a squared loss

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}$$

Symbols N number of readings; σ^2 common variance; residual $r = b - A\theta$.

Estimation

The familiar normal equation returns

$$A^T A\hat{\theta} = A^T b, \qquad A^T(b - A\hat{\theta}) = 0$$

Symbols $\hat{\theta}$ MLE; full column rank gives uniqueness; residual is orthogonal to columns.

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Symbols $\theta = (\text{intercept}, \text{slope})^T$; $r = b - A\hat{\theta}$.

WORKED EXAMPLE

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

Result $\hat{\theta}$ estimate; r observed minus fitted.

Estimation

A Gaussian prior sets a penalty scale

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \quad \lambda^2 = \sigma^2/\tau^2$$

Symbols $\theta \sim \mathcal{N}(\mu_0, \tau^2 I)$ prior; σ^2 noise variance; λ^2 relative strength.

Estimation

The prior changes the normal equation

$$(A^T A + \lambda^2 I)\hat{\theta} = A^T b + \lambda^2 \mu_0$$

Symbols $\lambda^2 > 0$ gives a unique MAP; use the previous fit with $\mu_0 = 0, \lambda^2 = 1$.

WORKED EXAMPLE

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \text{SSE} = \tfrac{2}{5}$$

Result $\hat{\theta}_{MAP}$ posterior mode; SSE squared residual.

CHOOSE, EXPLAIN, VOTE

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; prior center 0, $\lambda^2 = 1$. Why is MAP unique?

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

CHOOSE, EXPLAIN, VOTE

Integrated board

Calibration: choose the likelihood

$b = (1, 2, 3)^T$, $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model?

- **A** $\frac{1}{2}\sum_i (b_i - \theta)^2$
- **B** $\sum_i |b_i - \theta|$
- **C** $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

CHOOSE, EXPLAIN, VOTE

Integrated board

Calibration: combine data and prior

Same readings $(1, 2, 3)$; noise variance 1; prior $\theta \sim \mathcal{N}(0, 1)$.
How does the posterior compare with the MLE?

A. Center stays at 2; prior changes only uncertainty.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

CHOOSE, EXPLAIN, VOTE

Intervals

Calibration: confidence coverage

Same sensor: $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$ iid. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. What does 95% confidence mean?

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

CHOOSE, EXPLAIN, VOTE

Intervals

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability?

A. 95% posterior mass for θ, with b fixed.

B. 95% coverage for every fixed true offset.

C. 95% probability for a future noisy reading.

CHOOSE, EXPLAIN, VOTE

Intervals

Calibration: why do the intervals differ?

Same data: $C(b) = 2 \pm 1.96/\sqrt{3}$; $I(b) = 1.5 \pm 1.96(.5)$. Why is I shifted and narrower?

A. It must have better repeated-sampling coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

CHOOSE, EXPLAIN, VOTE

Transfer

Calibration: independent transfer

New readings $b = (2, 3)^T$; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which conclusion is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

CHOOSE, EXPLAIN, VOTE

Intervals

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

CHOOSE, EXPLAIN, VOTE

Synthesis

Which modeling bridge is valid?

With iid Gaussian noise and an independent isotropic Gaussian prior, which modeling chain is valid?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; independence gives sums.

C. A Gaussian prior removes the data residual term.

References

Sources and optional extensions

1. Textbook · Goodfellow et al., ch.3
2. Textbook · Goodfellow et al., §5.5–5.6
3. Course notes · UBC Gaussian, pp.5,7–8
4. NIST · Confidence intervals
5. Optional course · MIT probability